EDUSYNC ENTERPRISE AI

INFRASTRUCTURE PROPOSAL

Private. Powerful. Scalable.

Prepared for: Strategic Investors

Targeting: 100,000+ User Ecosystem

Date: March 9, 2026

Prepared for: Strategic Investors

Project: EduSync 4.0 - High-Performance Private AI Infrastructure

Date: March 10, 2026

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1. Executive Summary

EduSync Enterprise is a Tier-3 private AI infrastructure designed for institutions to own their intelligence. With a **Rs. 35 Lakhs investment**, we deploy the world's most powerful consumer-grade GPU cluster featuring **4x NVIDIA RTX 5090 (128GB VRAM)**. This system is capable of running state-of-the-art models like **Gemma 3 27B** and massive swarms of fine-tuned SLMs locally, serving over **1,00,000 users** with Zero Latency.

2. Vision and Mission

Establish a self-sustaining AI ecosystem where large-scale LLMs power every classroom, research lab, and administrative task with zero latency and 100% data sovereignty.

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3. The 128GB Beast: Hyper-Scaling Strategy

To support **1,00,000 - 1,50,000 users** on a single server, we leverage the massive bandwidth of the **RTX 50 series (Blackwell Architecture).**

The "Ultimate-Node" Tech Stack

- **Base Hardware**: Single High-Density Super-Node (**4x RTX 5090**).
- **Model Swarm**: 50+ Parallel Fine-Tuned 8B Models or a high-precision **Gemma 3 27B** anchor model.
- **VRAM Advantage**: 128GB of GDDR7 memory allows for massive "Continuous Batching," handling thousands of simultaneous user prompts without a queue.

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3. Target Audience

Our infrastructure is designed for high-density academic and professional environments:

- **Higher Education Institutions**: Universities and Engineering colleges requiring private research and learning tools.
- **K-12 School Networks**: Scaling AI tutors across multiple branches securely.
- **Corporate Training Hubs**: Enterprises training employees on internal proprietary data.
- **Government EdTech Initiatives**: Large-scale digital learning portals with strict data residency requirements.

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4. Hardware Architecture: The AI Super-Node

Itemized Hardware Audit (4x RTX 5090 Configuration)

Component	Specification	Estimated Cost (INR)
GPU Cluster	4x ASUS ROG Astral RTX 5090 (128GB VRAM)	Rs. 15,20,000
Processor	AMD Threadripper 7980X (64C / 128T)	Rs. 4,75,000
Motherboard	ASUS WRX90 Sage (PCIe 5.0 Workstation)	Rs. 1,15,000
Memory	512GB DDR5-5600 ECC Registered RAM	Rs. 2,90,000
Storage	8TB Gen5 NVMe RAID + 18TB Enterprise HDD	Rs. 90,000
Power Unit	Dual 2400W Titanium Redundant Server PSU	Rs. 1,50,000

Thermal System	Industrial Multi-GPU Custom Water Loop	Rs. 1,50,000
Chassis	4U Heavy-Duty Liquid-Ready Rack Chassis	Rs. 45,000
Networking	10Gbps SFP+ Fiber Core Connectivity	Rs. 30,000
Total Hardware	**Rs. 28,65,000**	

5. Data Center Infrastructure (Facility Requirements)

Running a high-performance cluster (3200W+ load) requires more than just a standard office room. Below are the infrastructure and associated costs to build a "Mini Data Center" locally.

Facility Setup Cost (Industrial Standard)

Building Capacity & Environment

Item	Specification	Estimated Cost (INR)
Power Backup	10kVA Online UPS with Extended Bank	Rs. 2,50,000
Dedicated Cooling	3-Ton Industrial Split-Inverter Hub	Rs. 1,50,000
Server Cabinet	12U Sound-Proof Thermal Server Rack	Rs. 15,000
Electrical Work	63A Industrial Line + Specialist Earthing	Rs. 1,20,000
Security & Safety	Novec-1230 Fire Suppression + Biometrics	Rs. 1,00,000
Total Facility Cost	**Rs. 6,35,000**	

- **Ventilation**: Needs a dust-free, humidity-controlled environment (RH 40-60%).

- **Connectivity**: Redundant 1Gbps Fiber lines with Static IPs.

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6. Deployment Procedure (Step-by-Step)

To successfully deploy this "Local Setu[p]," the following physical and digital workflow is followed:

Phase 1: Civil & Electrical (Week 1)

1. **Site Preparation**: Clear a 5ft x 5ft dedicated secure area.
2. **Earthing Audit**: Ensure neutral-to-earth voltage is <1V to protect the expensive RTX 4090 GPUs.
3. **Electrical Grid**: Install the 6kVA Online UPS and dedicated circuit breakers (MCBs).

Phase 2: Hardware Assembly (Week 2)

1. **Component Testing**: Individual stress tests for Threadripper and 4090s.
2. **Liquid Cooling Loop**: Custom loop installation to ensure GPUs stay below 70°C under 100% load.
3. **Rack Mounting**: Securing the 4U chassis into the server cabinet.

Phase 3: Software & AI Stack (Week 3)

1. **OS**: Ubuntu 24.04 LTS (Server Edition).
2. **Drivers**: NVIDIA CUDA 12.x + cuDNN installation.
3. **Orchestration**: Docker + Kubernetes for self-healing AI instances.
4. **Inference Engine**: Deploying **vLLM** or **Ollama** for high-throughput API endpoints.

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7. Model Strategy: The Gemma 3 Swarm

To leverage the **128GB VRAM**, we implement a hybrid "Master-Worker" model architecture for unprecedented scale.

VRAM Partitioning (128GB Strategy)

1. **The Master (Gemma 3 27B IT)**: Consumes **25GB VRAM** (Quantized). Acts as the primary logic and career preparation engine for high-complexity prompts.

2. **The Swarm (25x Gemma 3 4B IT)**: Consumes **75GB VRAM**. Each 4B instance handles thousands of basic student queries/chats simultaneously.

3. **The Result**: Total capacity for **4,000+ per-second concurrent prompts**, serving an active user base of 1,00,000+ with zero queue time.

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9. Financial Analysis & ROI

Capital Expenditure (Ultimate Setup)

Operational Cost (Monthly)

Category	Description	Cost (INR)
Core Hardware	4x RTX 5090 Server Infrastructure	Rs. 28,65,000
Industrial Facility	Power, Cooling, and Security Hub	Rs. 6,35,000
Total Capex	**Strategic Institutional Asset**	**Rs. 35,00,000**
Category	Description	Monthly Cost (INR)
Staffing	Dedicated AI/Systems Administrator	Rs. 1,20,000
Utilities	Electricity & Maintenance Reserves	Rs. 30,000
Total Opex	**Rs. 1,50,000**	

10. Marketing & Go-To-Market (GTM) Strategy

To reach the **100,000 user** milestone, we implement a multi-channel acquisition strategy.

Initial Marketing Budget (Growth Phase)

CAC (Cost per Acquisition) Targets

Channel	Purpose	Monthly Allocation (INR)
B2B Direct Sales	Commission for Institutional Sales Reps	Rs. 50,000
Professional (LinkedIn)	Targeted Ads for Principals & Educators	Rs. 30,000
Social Media (IG/YT)	Influencer partnerships (Tech/Study)	Rs. 40,000
Campus Events	AI Workshops & Hackathons sponsorship	Rs. 30,000
Total Growth Budget	**Rs. 1,50,000**	

- **Projected CAC**: Rs. 20 - Rs. 40 per user (Single payment for lifetime lead or low monthly recurring).

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11. Revenue Potential (1,00,000 User Milestone)

By utilizing the **Hyper-Efficiency Swarm** (12x Fine-tuned models on 1 node), we achieve massive profitability with zero additional hardware cost.

Category	Monthly (1,00,000 Users)	Annual Project

Gross Revenue (1L x Rs. 25)	Rs. 25,00,000	Rs. 3,00,00,000
Operational Costs	- Rs. 1,50,000	- Rs. 18,00,000
Annual Profit	**Rs. 23,50,000**	**Rs. 2,82,00,000**

Note: Since the hardware is fixed at Rs. 35L, the profit scales exponentially as the user base grows.

12. Strategic Advantage: Why Local Hosting?

The "Local Setup" Edge

Running the infrastructure locally instead of using Cloud APIs (OpenAI/Google) provides critical benefits:

1. **100% Data Privacy**: Institutional data never leaves the campus network. Total compliance with global data residency laws.

2. **Zero Latency Performance**: Blazing fast AI responses even with massive models, as it bypasses the internet lag and API rate limits.

3. **No Recurring API Costs**: Cloud AI costs scale with usage (pay-per-token). Local hosting has a fixed cost regardless of how many billions of tokens students generate.

4. **Infinite Customization**: Allows the institution to "Fine-Tune" models on their specific textbooks, research papers, and local curriculum.

5. **Offline Capability**: The core AI features remain functional even during internet outages or bandwidth throttling.

6. **Full Ownership**: The model weights, the hardware, and the data remain as permanent institutional assets.

Financial Outlook

- **Estimated Savings**: Replaces ~Rs. 3.5L+/month in Cloud AI API costs.
- **Payback Period**: ~10 months.

- **Academic Lead**: Puts the institution at the forefront of AI research globally.

Submitted by:

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